



Once-weekly semaglutide 7.2 mg in adults with obesity (STEP UP): a randomised, controlled, phase 3b trial

Sean Wharton, Paula Freitas, Jøran Hjelmæsæth, Maria Kabisch, Kristian Kandler, Ildiko Lingvay, Maria Quiroga, Julio Rosenstock, W Timothy Garvey, on behalf of the STEP UP trial group*

Summary

Background Once-weekly subcutaneous semaglutide 2.4 mg is approved for weight management in people with obesity and related complications; however, some individuals do not reach their therapeutic goals with this dose. We aimed to test the efficacy and safety of a higher dose of semaglutide (7.2 mg) in people with obesity.

Methods STEP UP was a phase 3b, randomised, double-blind, placebo-controlled and active-controlled trial conducted across 95 hospitals, specialist clinics, and medical centres in 11 countries in adults with BMI 30 kg/m² or greater, without diabetes. Participants were randomly assigned (5:1:1) to receive once-weekly subcutaneous semaglutide 7.2 mg, 2.4 mg, or placebo, with a lifestyle intervention, for 72 weeks. Coprimary endpoints were percentage change in bodyweight and the proportion of participants with a bodyweight reduction of 5% or greater for semaglutide 7.2 mg versus placebo from baseline to week 72 (treatment policy estimand). Confirmatory secondary endpoints were percentage change in bodyweight with 7.2 mg versus 2.4 mg, change in waist circumference (cm), and proportion of participants with bodyweight reductions of 10% or greater, 15% or greater, 20% or greater, and 25% or greater versus placebo, and 20% or greater and 25% or greater versus 2.4 mg. Safety was assessed in all participants who received at least one dose of the trial product. This trial is registered with ClinicalTrials.gov (NCT05646706) and is now completed.

Findings Between Jan 1, 2023, and Nov 26, 2024, 1407 participants were randomly assigned to semaglutide 7.2 mg (n=1005), semaglutide 2.4 mg (n=201), or placebo (n=201). 1037 (73.7%) of 1407 participants were female, the mean age was 47 (SD 12) years, mean bodyweight was 113.0 (24.1) kg, and mean BMI was 39.9 (7.1) kg/m². The mean change in bodyweight was greater with semaglutide 7.2 mg versus 2.4 mg (−18.7% [SE 0.4] vs −15.6% [0.7]; estimated treatment difference [ETD] −3.1% [95% CI −4.7 to −1.6]; p<0.0001) and with semaglutide 7.2 mg versus placebo (−18.7% [0.4] vs −3.9% [0.6]; −14.8% [−16.2 to −13.4]; p<0.0001). Participants in the semaglutide 7.2 mg group were more likely than those in the placebo group to reach bodyweight reductions of 5% or greater (odds ratio 12.1 [95% CI 8.3 to 17.6]; p<0.0001), 10% or greater (14.5 [9.6 to 21.9]; p<0.0001), 15% or greater (20.3 [11.2 to 36.8]; p<0.0001), 20% or greater (27.3 [10.9 to 68.0]; p<0.0001), and 25% or greater (127.4 [36.8 to 441.4]; p<0.0001); and more likely than those in the 2.4 mg group to reach bodyweight reductions of 20% or greater (1.8 [1.3 to 2.4]; p=0.0006) and 25% or greater (2.4 [1.6 to 3.5]; p<0.0001). Improvements in waist circumference were observed with semaglutide 7.2 mg versus placebo (ETD −11.7 cm [95% CI −13.0 to −10.4]; p<0.0001). Gastrointestinal adverse events were more common with semaglutide 7.2 mg (711 [70.8%] of 1004) versus 2.4 mg (123 [61.2%] of 201) or placebo (86 [42.8%] of 201), as was dysaesthesia (230 [22.9%] , 12 [6.0%], and one [0.5%], respectively). Serious adverse events were reported by 68 (6.8%) of 1004 participants with semaglutide 7.2 mg, 22 (10.9%) of 201 with semaglutide 2.4 mg, and 11 (5.5%) of 201 with placebo.

Interpretation Semaglutide 7.2 mg was superior to placebo and 2.4 mg for bodyweight reduction in adults with obesity, while retaining a favourable risk–benefit profile.

Funding Novo Nordisk.

Copyright © 2025 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Introduction

As the global prevalence of obesity approaches one billion people,¹ there is a need for treatment options that will allow people with obesity to reach their individual treatment goals to improve their health. Effective management of obesity requires options to reduce the risk of obesity-related complications and improve quality of life and wellbeing.^{2–5}

There is a continuously emerging landscape of effective obesity treatments, with many currently in development. Once-weekly subcutaneous semaglutide 2.4 mg, a GLP-1 receptor agonist, has been extensively investigated in the global phase 3 STEP clinical trial programme.^{6–16} In the STEP 1 trial, semaglutide 2.4 mg provided superior bodyweight reduction versus placebo, with a mean change of −14.9% versus −2.4%, respectively, after

Lancet Diabetes Endocrinol 2025; 13: 949–63

Published Online
September 14, 2025
[https://doi.org/10.1016/S2213-8587\(25\)00226-8](https://doi.org/10.1016/S2213-8587(25)00226-8)

See [Comment](#) page 901

See [Articles](#) page 935

For the German translation of the abstract see Online for appendix 1

For the Greek translation of the abstract see Online for appendix 2

*The complete list of STEP UP principal investigators is provided in appendix 3 (p 3)

Faculty of Medicine, University of Toronto, Toronto, ON, Canada (S Wharton MD); Wharton Faculty of Health and Kinesiology, York University, Toronto, ON, Canada (S Wharton); Faculty of Medicine, McMaster University, Hamilton, ON, Canada (S Wharton); Department of Medicine, Faculty of Medicine and i3S-Institute for Research and Innovation in Health, University of Porto, Porto, Portugal (P Freitas PhD); Integrated Responsibility Center for Obesity (CRI-O), São João Local Health Unit (ULS), Porto, Portugal (P Freitas); Department of Endocrinology, Obesity, and Nutrition, Vestfold Hospital Trust, Tønsberg, Norway (J Hjelmæsæth MD); Department of Endocrinology, Morbid Obesity, and Preventive Medicine, Institute of Clinical Medicine, University of Oslo, Oslo, Norway (J Hjelmæsæth); Novo Nordisk, Søborg, Denmark (M Kabisch PhD, K Kandler MD, M Quiroga MD); Department of Internal Medicine/Endocrinology and Peter O'Donnell Jr School of Public Health, University of Texas Southwestern Medical Center, Dallas, TX, USA

(Prof I Lingway MD); Velocity Clinical Research at Medical City, Dallas, TX, USA (J Rosenstock MD); Department of Nutrition Sciences, University of Alabama at Birmingham, Birmingham, AL, USA (Prof W T Garvey MD)

Correspondence to: Dr Sean Wharton, Wharton Medical Clinic Weight and Diabetes Management, Burlington, ON L7M 4Y1, Canada sean.wharton@utoronto.ca

See Online for appendix 3

Research in context

Evidence before this study

Although we did not conduct a systematic literature search before doing this study, evidence shows that semaglutide 2.4 mg is effective for weight management. However, some people did not reach their individual treatment goals with this dose in the STEP 1 trial in people with overweight or obesity. Modelling and simulation of semaglutide at different doses has suggested that a dose higher than the currently approved 2.4 mg might provide further bodyweight reduction and health benefits without jeopardising safety.

Added value of this study

To the best of our knowledge, this is the first randomised controlled trial to investigate the efficacy and safety of once-weekly semaglutide 7.2 mg in adults with obesity (BMI ≥ 30 kg/m²) without diabetes. After 72 weeks, the mean change in bodyweight was significantly greater with semaglutide 7.2 mg versus placebo (estimated difference -14.8%) and with semaglutide 7.2 mg versus semaglutide 2.4 mg (estimated difference -3.1%). Nearly half of participants had a bodyweight reduction of 20% or greater and almost a third had a bodyweight reduction of 25% or greater with semaglutide 7.2 mg. Other measures, including waist

circumference, blood pressure, HbA_{1c}, and fasting plasma glucose were reduced with semaglutide 7.2 mg versus placebo. A higher proportion (83.4%) of participants in the semaglutide 7.2 mg group with prediabetes at baseline reverted to normoglycaemia by week 72, compared with those in the semaglutide 2.4 mg (73.8%) and placebo (36.6%) groups. Adverse events associated with the drug class were more frequently reported with semaglutide 7.2 mg than with semaglutide 2.4 mg and placebo: namely, gastrointestinal disorders and dysaesthesia. Dysaesthesia events resolved before the end of the trial in most participants, although four participants in the semaglutide 7.2 mg group had to permanently discontinue due to these events, compared with no participants in the semaglutide 2.4 mg or placebo groups.

Implications of all the available evidence

These results support a favourable benefit-risk profile of semaglutide 7.2 mg for weight management in people with obesity. The greater magnitudes of bodyweight reduction seen in this trial, as well as the improvements in cardiovascular risk factors and glucose control, can have beneficial effects on health goals in clinical practice.

68 weeks (estimated treatment difference [ETD] -12.4% [95% CI -13.4 to -11.5]; $p < 0.0001$), and was well tolerated.⁶ Semaglutide 2.4 mg is approved for weight management and for the reduction of major adverse cardiovascular events for people with established cardiovascular disease and a BMI of 27 kg/m² or greater.^{17,18} The benefits of semaglutide 2.4 mg also include decreased pain for those with knee osteoarthritis and improved glycaemic status for those with prediabetes.^{14,15} Tirzepatide is a peptide engineered from the glucose-dependent insulinotropic polypeptide (GIP), investigated at once-weekly doses of 5 mg, 10 mg, and 15 mg.¹⁹ In the SURMOUNT-1 trial, all doses of tirzepatide provided superior bodyweight reduction versus placebo, with a mean change of -15.0% (ETD -11.9% [95% CI -13.4 to -10.4]; $p < 0.001$) with the 5 mg dose, -19.5% (-16.4% [-17.9 to -14.8]; $p < 0.001$) with the 10 mg dose, and -20.9% (-17.8% [-19.3 to -16.3]; $p < 0.001$) with the 15 mg dose versus -3.1% with placebo after 72 weeks, and all doses were well tolerated.¹⁹ In the SURMOUNT-5 trial, conducted in adults with overweight or obesity and at least one obesity-related complication, without type 2 diabetes, the mean percentage change in bodyweight was -13.7% (95% CI -14.9 to -12.6) with semaglutide 1.7 mg or 2.4 mg, and -20.2% (-21.4 to -19.1) with tirzepatide 10 mg or 15 mg after 72 weeks.²⁰

Despite the effectiveness of current treatments for weight management, some individuals will not reach the therapeutic goals that are associated with greater bodyweight reduction. It was deemed relevant to investigate a higher dose of semaglutide 7.2 mg to

establish whether this dose provides further benefits and can help those for whom the current treatment landscape is inadequate. Incremental reductions in bodyweight after the maximum benefits have been reached with a lower dose can positively affect these people's lives. Modelling and simulation of efficacy and exposure data from the phase 3a STEP (subcutaneous semaglutide 2.4 mg) and OASIS (oral semaglutide) programmes suggested that increasing the dose of semaglutide might provide further bodyweight reduction and health benefits without jeopardising safety or significantly increasing the risk of adverse events.^{6,7,21,22} In the OASIS 1 study, exposure to oral semaglutide tripled during the escalation from 14 mg to 25 mg, without leading to an increased rate of adverse events,²² and the exposure response in the STEP programme revealed that gastrointestinal adverse events plateau at higher exposures.^{6,7} As a result of these findings, it was deemed appropriate to investigate an increased dose of subcutaneous semaglutide without further additional intermediate increases in the titration schedule.

The aim of the STEP UP trial was to assess the efficacy and safety of once-weekly semaglutide 7.2 mg versus placebo, and also versus semaglutide 2.4 mg, as an adjunct to lifestyle intervention, in adults with obesity without type 2 diabetes.

Methods

Study design and participants

STEP UP was a phase 3b, randomised, double-blind, placebo-controlled and active-controlled, three-arm trial,

with a treatment duration of 72 weeks and a 9-week off-treatment follow-up. The trial was conducted across 95 hospitals, specialist clinics, and medical centres in 11 countries (Bulgaria, Canada, Germany, Greece, Hungary, Norway, Poland, Portugal, Slovakia, South Africa, and the USA). A list of the principal investigators is provided in appendix 3 (pp 3–5). The trial was conducted in accordance with the Declaration of Helsinki and the International Conference on Harmonization of Technical Requirements for Pharmaceuticals for Human Use Good Clinical Practice guidelines. The protocol and amendments were approved by the relevant institutional review board or independent ethics committee at each trial site and are available in appendix 3 (p 39).

All participants provided written informed consent before randomisation. Participants aged 18 years or older at the time of consent with a BMI of 30 kg/m² or greater were eligible and were required to have a history of at least one self-reported unsuccessful dietary effort to lose bodyweight. Sex was reported by the investigator or physician as female or male in the electronic health record at baseline. Race and ethnicity were self-reported by participants. Key exclusion criteria included a history of type 1 diabetes or type 2 diabetes, HbA_{1c} 6·5% or greater (48 mmol/mol), and a self-reported change in bodyweight greater than 5 kg within 90 days before screening. A full list of eligibility criteria is provided in appendix 3 (pp 6–7).

This trial is registered with (NCT05646706) and is now completed.

Randomisation and masking

Participants were randomly assigned in a 5:1:1 ratio to receive once-weekly subcutaneous semaglutide 7·2 mg, semaglutide 2·4 mg, or placebo (appendix 3 p 26), to ensure adequate exposure to the 7·2 mg dose primarily under investigation. Participants were randomly assigned with the Randomization and Trial Supplies Management System or Interactive Web Response System, in a double-blind manner. Randomisation was done with a block size of 10, with no stratification factors. All groups received visually identical injections to maintain masking of participants and site staff.

Procedures

Participants followed a fixed dose escalation regimen. Semaglutide was initiated at 0·25 mg, increasing every 4 weeks to 2·4 mg at week 16. At week 20, participants started on their respective maintenance doses of 7·2 mg, 2·4 mg, or placebo. Participants in the placebo group continued on placebo, participants in the 2·4 mg group remained on 2·4 mg from week 16, and at week 20, participants in the semaglutide 7·2 mg group directly escalated to this dose without additional titration. A delay to dose escalation, or dose reduction, was allowed if the participant was unable to tolerate the 2·4 mg or 7·2 mg maintenance dose and would otherwise discontinue

treatment. It was recommended that each participant made at least one attempt to re-escalate, before staying on a lower dose. Participants were encouraged to remain in the trial regardless of whether they had to discontinue the trial product permanently. Participants also underwent a lifestyle intervention comprising a reduced calorie diet and increased physical activity (–500 kcal per day and 150 min per week of physical activity). The lifestyle intervention was monitored by the dietitian as part of regular counselling, but no data relating to adherence were collected.

Following the dose-escalation period, there was a maintenance period of 52 weeks, during which participants received their respective target doses. Off-treatment follow-up was 9 weeks. Participants were advised not to initiate any other obesity therapies. If these were initiated, participants were instructed to stop these.

A schedule of assessments is provided in appendix 3 (pp 12–13).

Outcomes

All efficacy endpoints were assessed from baseline to week 72, with safety endpoints assessed up to week 81. The coprimary endpoints were percentage change in bodyweight and the proportion of participants with a bodyweight reduction of 5% or greater for semaglutide 7·2 mg versus placebo, in line with regulatory requirements.^{23,24} The confirmatory secondary endpoints were percentage change in bodyweight for semaglutide 7·2 mg versus semaglutide 2·4 mg; change in waist circumference (cm) for semaglutide 7·2 mg versus placebo; and the proportion of participants with a bodyweight reduction of 10% or greater, 15% or greater, 20% or greater, and 25% or greater with semaglutide 7·2 mg versus placebo, and 20% or greater and 25% or greater with semaglutide 7·2 mg versus semaglutide 2·4 mg. Coprimary and confirmatory secondary endpoints were predefined in a statistical hierarchy (appendix 3 pp 8–9) and had enough power to infer superiority.

The supportive secondary endpoints were change in absolute bodyweight (kg), BMI (kg/m²), systolic and diastolic blood pressure (mm Hg), HbA_{1c} (percentage points), fasting plasma glucose (mmol/L), and glycaemic category (count of participants with normoglycaemia, prediabetes, or incident type 2 diabetes); and ratio to baseline in lipids (triglycerides, total cholesterol, high-density lipoprotein [HDL] cholesterol, low-density lipoprotein [LDL] cholesterol, and very low-density lipoprotein [VLDL] cholesterol; mmol/mol), high-sensitivity C-reactive protein (hsCRP), and fasting serum insulin; all for semaglutide 7·2 mg versus placebo. Glycaemic category was based on all relevant information, including medical records, concomitant medication, blood glucose parameters, and adverse events, and was categorised by the investigator as normoglycaemia, prediabetes, or incident type 2 diabetes

according to the American Diabetes Association's definitions.

Exploratory analyses included change in the number of repetitions in the 30-s sit-to-stand test and the proportion of participants with a BMI less than 27 kg/m² at week 72 with semaglutide 7.2 mg versus placebo (appendix 3 p 11). The proportion of participants with a BMI less than 25 kg/m² and less than 30 kg/m² at week 72, as well as those with a waist-to-height ratio less than 0.5, was also explored as a post-hoc exploratory analysis for semaglutide 7.2 mg versus placebo. All endpoints that were not prespecified to compare

semaglutide 7.2 mg with semaglutide 2.4 mg were analysed post hoc.

A subset of participants also underwent an MRI scan at randomisation and end of treatment, to assess changes in total, lean, and visceral fat volume (L) in the semaglutide (pooled dose) and placebo groups. MRI was performed at sites in Canada, Greece, South Africa, and the USA. Participants were included in the MRI subset during enrolment at those sites, provided they met specific eligibility criteria.

Safety was assessed by monitoring adverse events (adverse events were not reviewed by an independent adjudication committee). Dysaesthesia-associated adverse events included the following conditions: allodynia, burning sensation, dysaesthesia, hyperaesthesia, pain of skin, paraesthesia, sensitive skin, skin burning sensation, skin discomfort, and skin sensitisation. As a result of dysaesthesia adverse events being identified in other semaglutide trials,¹⁷ the informed consent forms were updated on June 27, 2023, to include dysaesthesia adverse events as per a protocol amendment; the investigator's brochure was also updated. Reconsent was required from all participants, unless they were approaching the end-of-treatment visit.

The full list of endpoints is provided in appendix 3 (pp 10–11).

Statistical analysis

A sample size of 1400 participants (1000 in the semaglutide 7.2 mg group, 200 in the semaglutide 2.4 mg group, and 200 in the placebo group) was estimated to provide an effective power of 84% for the coprimary and confirmatory secondary endpoints, tested at a significance level of 5% (two-sided) in the prespecified statistical hierarchy (appendix 3 pp 8–9) to show the superiority of semaglutide 7.2 mg versus semaglutide 2.4 mg (based on confirmatory secondary endpoints) or placebo (based on coprimary endpoints and confirmatory secondary endpoints). All endpoints in the statistical hierarchy were tested at the same local significance level alpha ($\alpha=0.05$), moving to the next endpoint only after a success on the previous endpoint, and were therefore controlled for type 1 error. All treatment differences are accompanied by a two-sided 95% CI and a p value.

Efficacy endpoints were analysed in the full analysis set, which included all randomly assigned participants. Safety endpoints were analysed in the safety analysis set, which included all participants who received at least one dose of the trial product.

Two observation periods were defined (appendix 3 p 9). Efficacy assessments from the in-trial period were analysed according to the treatment policy (primary) estimand, which assessed treatment effect among all randomly assigned participants, regardless of trial product discontinuation, dose reduction, or initiation of any other obesity therapies. Continuous endpoints were analysed with an analysis of covariance model (linear

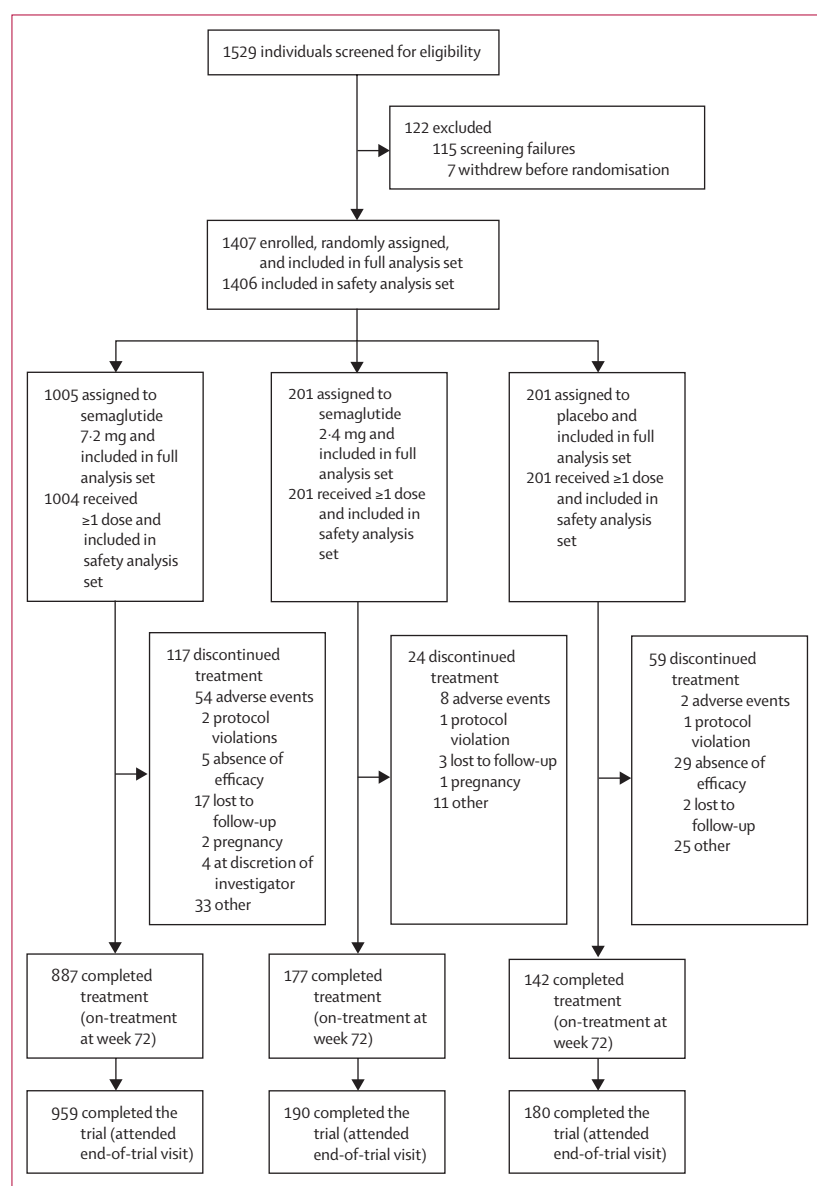


Figure 1: Trial profile

During the trial, 14 (1.4%) of 1005 participants in the semaglutide 7.2 mg group, three (1.5%) of 201 in the semaglutide 2.4 mg group, and 18 (9.0%) of 201 in the placebo group initiated another obesity medication or had bariatric surgery, or both.

regression) with randomised treatment as a factor and the baseline value of the endpoint as a covariate. Categorical endpoints were analysed by use of a logistic regression model with randomised treatment as a factor and the baseline value of the endpoint as a covariate. A multiple imputation approach was used to impute missing data 1000 times with assessments from retrieved participants in the same treatment group, to account for uncertainty, with similarities to the approach described by McEvoy²⁵ and previously used in phase 3 weight management trials.^{19,26} The datasets of 1000 results were combined into one set with Rubin's rule. Missing measurements at week 72 were imputed from available measurements at the same time point (week 72).

Assessments from the on-treatment period until first treatment discontinuation were analysed according to the trial product (secondary) estimand, which assessed treatment effect in randomly assigned participants as though all participants had remained on treatment for the duration of the trial and had not initiated other obesity therapies. Continuous endpoints were analysed with a mixed model for repeated measures analysis, with randomised treatment as a factor and the baseline value of each endpoint as a covariate, all nested within visit as a factor. **Categorical endpoints were analysed by use of logistic regression, with randomised treatment as a factor and the baseline value of the endpoint as a covariate.** For categorical endpoints, missing data at week 72 will be predicted from the mixed model for repeated measures analysis.

Efficacy endpoints were analysed up to week 72 (end-of-treatment visit), while safety endpoints were analysed up to week 81 (end-of-trial visit). Statistical analyses were done with SAS software (version 9.4). Presented efficacy data are for the treatment policy estimand, unless otherwise stated.

Role of the funding source

The funder provided the trial product, designed the study, oversaw its conduct, monitored study sites, and collected and analysed the data; investigators were responsible for study-related medical decisions and data collection. This manuscript was drafted under the guidance of the authors, with medical writing and editorial support paid for by the funder.

Results

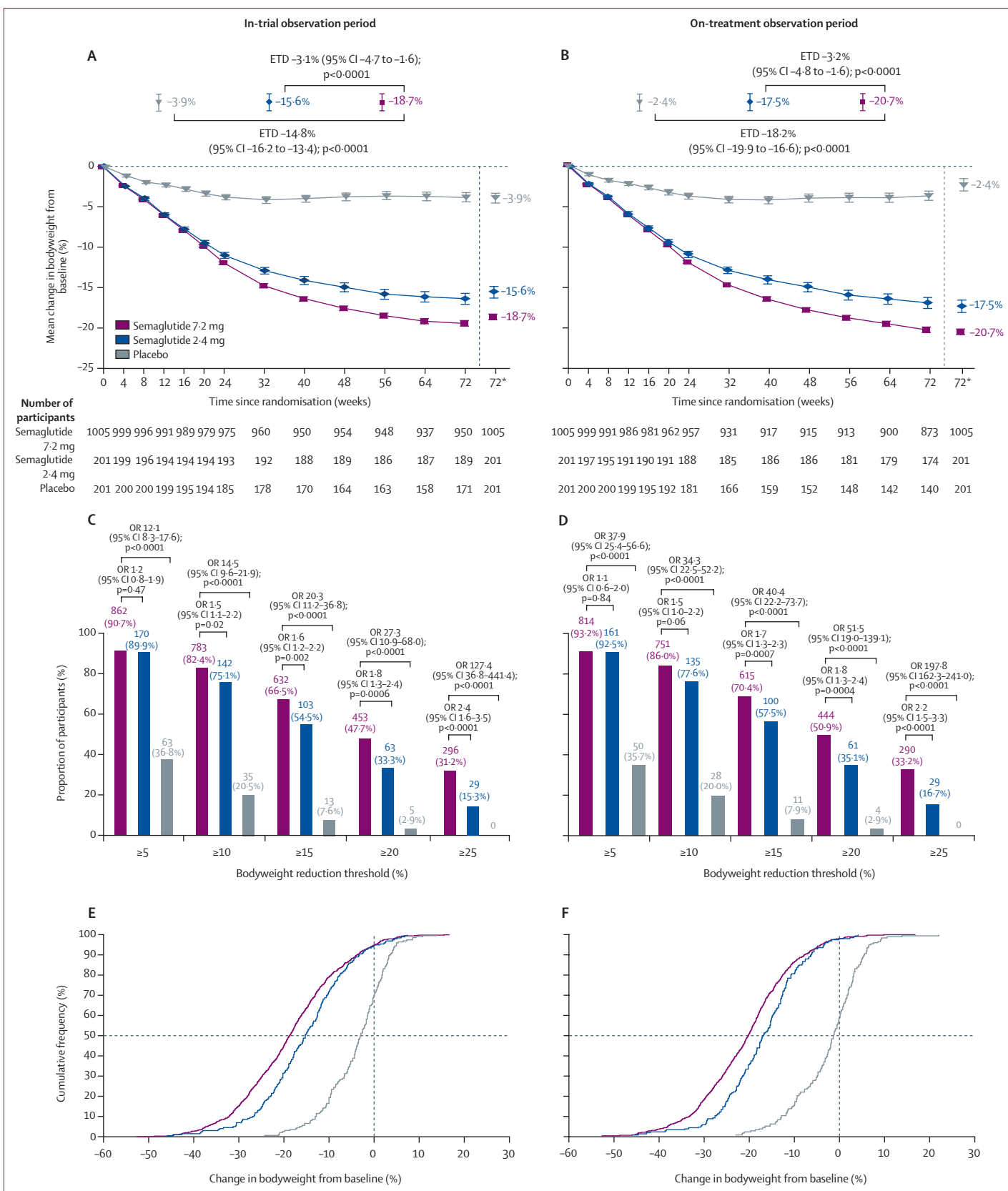
Between Jan 1, 2023, and Nov 26, 2024, 1529 participants were screened, of whom 1407 were randomly assigned to receive semaglutide 7.2 mg (n=1005), semaglutide 2.4 mg (n=201), or placebo (n=201) and included in the full analysis set (figure 1). Most participants completed the trial (attended end-of-trial visit at week 81; 959 [95.4%] of 1005 on semaglutide 7.2 mg, 190 [94.5%] of 201 on semaglutide 2.4 mg, and 180 [89.6%] of 201 on placebo), and most participants

completed the trial on treatment at week 72 (887 [88.3%] on semaglutide 7.2 mg, 177 [88.1%] on semaglutide 2.4 mg, and 142 [70.6%] on placebo). Among the treatment completers, 669 (75.4%) of 887 participants in the semaglutide 7.2 mg group, 158 (89.3%) of 177 in

	Semaglutide 7.2 mg (n=1005)	Semaglutide 2.4 mg (n=201)	Placebo (n=201)	Total (N=1407)
Sex				
Female	753 (74.9%)	137 (68.2%)	147 (73.1%)	1037 (73.7%)
Male	252 (25.1%)	64 (31.8%)	54 (26.9%)	370 (26.3%)
Race				
White	859 (85.5%)	176 (87.6%)	168 (83.6%)	1203 (85.5%)
Black or African American	82 (8.2%)	17 (8.5%)	22 (10.9%)	121 (8.6%)
Asian	52 (5.2%)	4 (2.0%)	7 (3.5%)	63 (4.5%)
Other*	10 (1.0%)	3 (1.5%)	4 (2.0%)	17 (1.2%)
Missing	2 (0.2%)	1 (0.5%)	0	3 (0.2%)
Ethnicity				
Hispanic or Latino	44 (4.4%)	12 (6.0%)	7 (3.5%)	63 (4.5%)
Not Hispanic or Latino	961 (95.6%)	189 (94.0%)	194 (96.5%)	1344 (95.5%)
Age, years	47 (12)	46 (12)	48 (12)	47 (12)
Bodyweight, kg	112.4 (23.8)	116.5 (26.2)	112.4 (22.8)	113.0 (24.1)
BMI, kg/m ²	39.8 (7.0)	40.5 (7.8)	39.7 (6.6)	39.9 (7.1)
Waist circumference, cm	118.4 (15.8)	120.3 (16.9)	118.6 (14.5)	118.7 (15.8)
Waist to height ratio	0.7 (0.1)	0.7 (0.1)	0.7 (0.1)	0.7 (0.1)
HbA _{1c} , %	5.7% (0.3)	5.6% (0.4)	5.7% (0.3)	5.7% (0.3)
HbA _{1c} , mmol/mol	38.4 (3.7)	38.0 (3.9)	38.6 (3.8)	38.4 (3.7)
Glycaemic status at baseline†				
Normoglycaemia	627 (62.4%)	132 (65.7%)	118 (58.7%)	877 (62.3%)
Prediabetes	378 (37.6%)	69 (34.3%)	82 (40.8%)	529 (37.6%)
eGFR, mL/min per 1.73 m ² ‡	95.1 (19.6)	96.4 (18.0)	95.5 (20.4)	95.3 (19.5)
Obesity-related complications in >5% of participants overall at screening				
Hypertension	431 (42.9%)	75 (37.3%)	79 (39.3%)	585 (41.6%)
Dyslipidaemia	277 (27.6%)	55 (27.4%)	60 (29.9%)	392 (27.9%)
Gastro-oesophageal reflux disease	131 (13.0%)	31 (15.4%)	27 (13.4%)	189 (13.4%)
Cardiovascular disease	107 (10.6%)	21 (10.4%)	23 (11.4%)	151 (10.7%)
Knee osteoarthritis	60 (6.0%)	20 (10.0%)	11 (5.5%)	91 (6.5%)
Liver disease	79 (7.9%)	16 (8.0%)	15 (7.5%)	110 (7.8%)
Asthma	92 (9.2%)	12 (6.0%)	23 (11.4%)	127 (9.0%)
Depression	96 (9.6%)	14 (7.0%)	19 (9.5%)	129 (9.2%)
Anxiety	88 (8.8%)	16 (8.0%)	24 (11.9%)	128 (9.1%)
Obstructive sleep apnoea	68 (6.8%)	24 (11.9%)	15 (7.5%)	107 (7.6%)
Number of obesity-related complications at screening				
None	328 (32.6%)	71 (35.3%)	65 (32.3%)	464 (33.0%)
1	274 (27.3%)	48 (23.9%)	59 (29.4%)	381 (27.1%)
2	202 (20.1%)	33 (16.4%)	33 (16.4%)	268 (19.0%)
3	102 (10.1%)	20 (10.0%)	25 (12.4%)	147 (10.4%)
4	55 (5.5%)	17 (8.5%)	13 (6.5%)	85 (6.0%)
≥5	44 (4.4%)	12 (6.0%)	6 (3.0%)	62 (4.4%)

Data are n (%) or mean (SD). Obesity-related complications data were collected from the self-reported patient medical histories. eGFR=estimated glomerular filtration rate. *Other comprises the following races: American Indian or Alaska Native, Native Hawaiian, Other Pacific Islander, or multiple ethnicities. †Evaluated by the Investigator based on all available or relevant information. ‡Geometric mean (coefficient of variation).

Table 1: Baseline characteristics



the semaglutide 2.4 mg group, and 137 (96.5%) of 142 in the placebo group finished the trial on the maximum allocated dose at week 72 (appendix 3 p 27).

Baseline characteristics were balanced across groups (table 1). Among the 1407 participants, most were female (1037 [73.7%]) and White (1203 [85.5%]). The mean age was 47 (SD 12) years, mean bodyweight was 113.0 (24.1) kg, and mean BMI was 39.9 (7.1) kg/m². Overall, 529 (37.6%) participants had prediabetes. The most common obesity-related complications at screening were hypertension (585 [41.6%]) and dyslipidaemia (392 [27.9%]). Anxiety was present in 128 (9.1%) of 1407 participants and depression was present in 129 (9.2%), while 261 (18.6%) participants were receiving psychoactive medication at randomisation (appendix 3 p 14).

Using the treatment policy estimand, mean change in bodyweight from baseline to week 72 was significantly greater with semaglutide 7.2 mg (−18.7% [SE 0.4]) than with semaglutide 2.4 mg (−15.6% [0.7]) and placebo (−3.9% [0.6]; figure 2A). The ETD for semaglutide 7.2 mg versus semaglutide 2.4 mg was −3.1% (95% CI −4.7 to −1.6; $p<0.0001$) and that for semaglutide 7.2 mg versus placebo was −14.8% (−16.2 to −13.4; $p<0.0001$; table 2). Using the trial product estimand, the mean change in bodyweight was −20.7% (SE 0.3) with semaglutide 7.2 mg, −17.5% (0.7) with semaglutide 2.4 mg, and −2.4% (0.8) with placebo (figure 2B). The corresponding ETD for semaglutide 7.2 mg versus semaglutide 2.4 mg was −3.2% (95% CI −4.8 to −1.6; $p<0.0001$), and that for placebo was −18.2% (−19.9 to −16.6; $p<0.0001$; figure 2B).

Participants in the semaglutide 7.2 mg group were significantly more likely than those in the placebo groups to reach a bodyweight reduction of 5% or greater (odds ratio [OR] 12.1 [95% CI 8.3 to 17.6]; $p<0.0001$), and more likely to reach a bodyweight reduction of 10% or greater, 15% or greater, 20% or greater, and 25% or greater versus placebo (table 2); and more likely than those in the semaglutide 2.4 mg group to reach a bodyweight reduction of 15% or greater, 20% or greater, and 25% or greater from baseline at week 72 (figure 2C; table 2). Nearly half of participants (453 [47.7%] of 950) in the

semaglutide 7.2 mg group reached a bodyweight reduction of 20% or greater at week 72, compared with a third (63 [33.3%] of 189) in the semaglutide 2.4 mg group and five (2.9%) of 171 in the placebo group. Additionally, 296 (31.2%) of 950 participants reached a bodyweight reduction of 25% or greater with semaglutide 7.2 mg, compared with 29 (15.3%) of 189 with semaglutide 2.4 mg, and none in the placebo group. Similar results were observed for the trial product estimand (figure 2D).

The cumulative change in bodyweight over the trial for both estimands is presented in figures 2E and 2F. Waterfall plots for the observed percentage bodyweight reduction reached by individual participants from baseline to week 72 for the on-treatment period are shown in appendix 3 (p 28).

The magnitude of bodyweight reductions in all treatment groups was numerically higher in women than in men (appendix 3 p 29). Age, BMI, geographical region, and glycaemic status did not affect the magnitude of bodyweight reductions or the proportion of participants with bodyweight loss of 5% or greater for semaglutide 7.2 mg versus 2.4 mg and placebo (appendix 3 p 16).

When applying the treatment policy estimand, significant reductions in waist circumference (figure 3), absolute bodyweight, and BMI were seen from baseline to week 72 with semaglutide 7.2 mg compared with placebo (table 2). When applying the trial product estimand, similarly significant reductions were seen for waist circumference (ETD −14.1 cm [95% CI −15.7 to −12.4]; $p<0.0001$), absolute bodyweight (−20.0 kg [−21.9 to −18.2]; $p<0.0001$), and BMI −7.2 kg/m² [−7.8 to −6.5]; $p<0.0001$) for semaglutide 7.2 mg versus placebo (appendix 3 pp 17–19). Reductions seen with semaglutide 7.2 mg versus semaglutide 2.4 mg were of a similar magnitude with both estimands (table 2; appendix 3 pp 17–19).

When comparing with semaglutide 7.2 mg with placebo, a greater proportion of participants reached BMI less than 25 kg/m², less than 27 kg/m², and less than 30 kg/m² at week 72 (table 2; appendix 3 p 30). When comparing semaglutide 7.2 mg with semaglutide 2.4 mg, a greater proportion of participants reached BMI less than 25 kg/m², less than 27 kg/m², and less than 30 kg/m² at week 72 (table 2; appendix 3 p 30).

A greater proportion of participants had a waist-to-height ratio less than 0.5 at week 72 with semaglutide 7.2 mg compared with semaglutide 2.4 mg and placebo (table 2).

Overall body composition improved with pooled semaglutide compared with placebo, with predominant reductions in fat versus lean body volume (appendix 3 p 31). Total fat volume changed by −25.1% with pooled semaglutide versus −0.9% with placebo (ETD −24.3% [95% CI −34.7 to −13.9]; $p<0.0001$), and lean body volume changed by −6.7% with pooled semaglutide versus 0.2% with placebo (−7.0% [−16.2 to 2.2]; $p=0.14$).

Figure 2: Change in bodyweight and proportion of participants with different thresholds of bodyweight reduction at week 72

Data are for the full analysis set. Treatment comparisons were estimated with the treatment policy estimand in panels A and C, and the trial product estimand in panels B and D. In panels A and B, error bars are standard error of the mean, and the numbers below the graphs are the number of participants contributing to the mean. Comparisons of semaglutide 7.2 mg versus 2.4 mg reported in this figure for the proportion of participants with bodyweight reductions of 5% or greater, 10% or greater, and 15% or greater were conducted in a post-hoc manner. Post-hoc analyses were not controlled for multiplicity, and findings for these endpoints should not be used to infer definitive treatment effects. Sections of this figure have been reproduced from Wharton et al²⁷ by permission of the American Diabetes Association. ETD=estimated treatment difference. OR=odds ratio. *Estimated mean change at week 72.

	Semaglutide 7·2 mg (n=1005)	Semaglutide 2·4 mg (n=201)	Placebo (n=201)	Treatment comparison: semaglutide 7·2 mg vs specified comparator (95% CI)*
Coprimary endpoints†				
Change in bodyweight, %	-18·7% (0·4); n=950	-15·6% (0·7); n=189	-3·9% (0·6); n=171	ETD vs 2·4 mg: -3·1% (-4·7 to -1·6); p<0·0001 ETD vs placebo: -14·8% (-16·2 to -13·4); p<0·0001
Proportion of participants with bodyweight reduction ≥5%	862/950 (90·7%)	170/189 (89·9%)	63/171 (36·8%)	OR vs 2·4 mg: 1·2 (0·8 to 2·0); p=0·47 OR vs placebo: 12·1 (8·3 to 17·6); p<0·0001
Confirmatory secondary endpoints				
Proportion of participants with bodyweight reduction ≥10%	783/950 (82·4%)	142/189 (75·1%)	35/171 (20·5%)	OR vs 2·4 mg: 1·5 (1·1 to 2·2); p=0·02 OR vs placebo: 14·5 (9·6 to 21·9); p<0·0001
≥15%	632/950 (66·5%)	103/189 (54·5%)	13/171 (7·6%)	OR vs 2·4 mg: 1·6 (1·2 to 2·2); p=0·002 OR vs placebo: 20·3 (11·2 to 36·8); p<0·0001
≥20%	453/950 (47·7%)	63/189 (33·3%)	5/171 (2·9%)	OR vs 2·4 mg: 1·8 (1·3 to 2·4); p=0·0006 OR vs placebo: 27·3 (10·9 to 68·0); p<0·0001
≥25%	296/950 (31·2%)	29/189 (15·3%)	0/171	OR vs 2·4 mg: 2·4 (1·6 to 3·5); p<0·0001 OR vs placebo: 127·4 (36·8 to 441·4); p<0·0001
Waist circumference, cm	-17·5 (0·3); n=950	-14·6 (0·8); n=189	-5·9 (0·6); n=171	ETD vs 2·4 mg: -2·9 (-4·6 to -1·3); p=0·0004 ETD vs placebo: -11·7 (-13·0 to -10·4); p<0·0001
Supportive secondary endpoints				
Absolute bodyweight, kg	-20·9 (0·4); n=950	-17·3 (0·8); n=189	-4·6 (0·7); n=171	ETD vs 2·4 mg: -3·6 (-5·3 to -1·8); p<0·0001 ETD vs placebo: -16·3 (-17·9 to -14·8); p<0·0001
BMI, kg/m ²	-7·5 (0·1); n=950	-6·1 (0·3); n=189	-1·6 (0·2); n=171	ETD vs 2·4 mg: -1·3 (-2·0 to -0·7); p<0·0001 ETD vs placebo: -5·8 (-6·4 to -5·3); p<0·0001
Blood pressure, mm Hg				
Systolic	-9·9 (0·4); n=949	-8·7 (1·0); n=188	-4·4 (0·8); n=171	ETD vs 2·4 mg: -1·2 (-3·4 to 1·0); p=0·29 ETD vs placebo: -5·4 (-7·3 to -3·6); p<0·0001
Diastolic	-4·5 (0·3); n=949	-3·0 (0·6); n=188	-2·1 (0·7); n=171	ETD vs 2·4 mg: -1·5 (-2·9 to -0·1); p=0·03 ETD vs placebo: -2·4 (-3·8 to -1·1); p=0·0006
Lipid concentrations, ratio to baseline, mmol/L				
Total cholesterol	0·95; n=941	0·95; n=187	1·00; n=168	ETR vs 2·4 mg: 1·00 (0·97 to 1·02); p=0·79 ETR vs placebo: 0·95 (0·90 to 1·00); p=0·06
HDL cholesterol	1·08; n=940	1·06; n=184	1·02; n=168	ETR vs 2·4 mg: 1·02 (1·00 to 1·05); p=0·04 ETR vs placebo: 1·06 (0·96 to 1·17); p=0·23
LDL cholesterol	0·93; n=940	0·93; n=184	0·99; n=168	ETR vs 2·4 mg: 1·00 (0·96 to 1·04); p=0·98 ETR vs placebo: 0·95 (0·89 to 1·00); p=0·05
VLDL cholesterol	0·77; n=941	0·85; n=186	0·99; n=168	ETR vs 2·4 mg: 0·91 (0·85 to 0·98); p=0·01 ETR vs placebo: 0·78 (0·72 to 0·84); p<0·0001
Triglycerides	0·77; n=941	0·84; n=186	0·99; n=168	ETR vs 2·4 mg: 0·92 (0·86 to 0·97); p=0·01 ETR vs placebo: 0·78 (0·72 to 0·84); p<0·0001
hsCRP, ratio to baseline, mg/L	0·40; n=942	0·42; n=187	0·90; n=168	ETR vs 2·4 mg: 0·94 (0·79 to 1·12); p=0·49 ETR vs placebo: 0·44 (0·37 to 0·53); p<0·0001
HbA _{1c} , percentage points	-0·32 (0·01); n=938	-0·30 (0·02); n=186	-0·02 (0·03); n=167	ETD vs 2·4 mg: -0·03 (-0·07 to 0·02); p=0·21 ETD vs placebo: -0·30 (-0·36 to -0·24); p<0·0001
Fasting plasma glucose, mmol/L	-0·63 (0·02); n=938	-0·56 (0·04); n=185	-0·12 (0·06); n=166	ETD vs 2·4 mg: -0·07 (-0·15 to 0·00); p=0·06 ETD vs placebo: -0·51 (-0·63 to -0·40); p<0·0001
Fasting serum insulin, ratio to baseline, pmol/L	0·61; n=921	0·68; n=184	0·88; n=163	ETR vs 2·4 mg: 0·89 (0·81 to 0·97); p=0·01 ETR vs placebo: 0·69 (0·62 to 0·77); p<0·0001
Participants with prediabetes at baseline and normoglycaemia at week 72	297/356 (83·4%)	48/65 (73·8%)	26/71 (36·6%)	..
Participants with prediabetes at baseline and incident type 2 diabetes at week 72	0/356	1/65 (1·5%)	3/71 (4·2%)	..
Decrease in antihypertensive medication	57/953 (6·0%)	10/190 (5·3%)	6/177 (3·4%)	..
Decrease in lipid-lowering medication	8/953 (0·8%)	1/190 (0·5%)	0/177	..

(Table 2 continues on next page)

	Semaglutide 7·2 mg (n=1005)	Semaglutide 2·4 mg (n=201)	Placebo (n=201)	Treatment comparison: semaglutide 7·2 mg vs specified comparator (95% CI)*
(Continued from previous page)				
Exploratory endpoints				
Participants with BMI <27 kg/m ² at week 72	253/950 (26·6%)	35/189 (18·5%)	1/171 (0·6%)	OR vs 2·4 mg: 1·6 (1·1 to 2·2); p=0·01 OR vs placebo: 36·9 (5·7 to 237·3); p=0·0001
Change in the sit-to-stand test at week 72, repetitions	2·73; n=941	2·82; n=185	2·15; n=166	ETD vs 2·4 mg: -0·1 (-0·7 to 0·5); p=0·77 ETD vs placebo: 0·6 (-0·2 to 1·3); p=0·14
Post-hoc analyses				
Participants with BMI <25 kg/m ² at week 72	148/950 (15·6%)	15/189 (7·9%)	0/171	OR vs 2·4 mg: 2·1 (1·2 to 3·4); p=0·006 OR vs placebo: 51·4 (14·3 to 184·2); p<0·0001
Participants with BMI <30 kg/m ² at week 72	410/950 (43·2%)	68/189 (36·0%)	14/171 (8·2%)	OR vs 2·4 mg: 1·3 (1·0 to 1·7); p=0·06 OR vs placebo: 7·2 (4·4 to 11·7); p<0·0001
Participants with waist-to- height ratio <0·5 at week 72	145/950 (15·3%)	13/189 (6·9%)	2/171 (1·2%)	OR vs 2·4 mg: 2·5 (1·4 to 4·3); p=0·001 OR vs placebo: 15·1 (3·8 to 59·9); p=0·0001
Data are for the full analysis set and are from the in-trial observation period. Data show the estimated mean (SE) change in each variable from baseline to week 72 or n/N (%) at week 72, unless otherwise stated. Treatment comparisons were estimated with the treatment policy estimand. Among treatment completers (semaglutide 7·2 mg, n=887; semaglutide 2·4 mg, n=177; placebo, n=142), 75·4% of participants randomly assigned to semaglutide 7·2 mg reached the final dose of 7·2 mg, 89·3% randomly assigned to semaglutide 2·4 mg reached the final dose of 2·4 mg, and 96·5% randomly assigned to placebo reached the equivalent of the final dose of 7·2 mg at week 72. ETD=estimated treatment difference. ETR=estimated treatment ratio. HDL=high-density lipoprotein. hsCRP=high-sensitivity C-reactive protein. LDL=low-density lipoprotein. OR=odds ratio. VLDL=very low-density lipoprotein. *Semaglutide 7·2 mg versus 2·4 mg comparisons reported in this table, except for change in bodyweight (%) and proportion of participants with bodyweight 20% or greater and 25% or greater were post hoc. Post-hoc analyses were not controlled for multiplicity, and findings for these endpoints should not be used to infer definitive treatment effects. †Coprimary endpoints were for semaglutide 7·2 mg versus placebo only. ‡Categorised as normoglycaemia (HbA _{1c} <5·7%), prediabetes (≥5·7% to <6·5%), or type 2 diabetes (HbA _{1c} ≥6·5%) according to the American Diabetes Association's definitions.				
Table 2: Coprimary, secondary, and selected exploratory trial endpoints from baseline to week 72				

Significant improvements in blood pressure and hsCRP were seen from baseline to week 72 with semaglutide 7·2 mg versus placebo (table 2). A reduction in diastolic blood pressure was observed between semaglutide 7·2 mg and semaglutide 2·4 mg; results were comparable between semaglutide 7·2 mg and semaglutide 2·4 mg for systolic blood pressure and hsCRP. Improvements in lipid concentrations were also seen with semaglutide 7·2 mg versus semaglutide 2·4 mg and placebo, including improvements in VLDL cholesterol and triglycerides, as well as LDL cholesterol versus placebo only (table 2). Change over time graphs for blood pressure are shown in appendix 3 (p 32).

Data with the trial product estimand are shown in appendix 3 (pp 17–19). When applying the treatment policy estimand, significant improvements in glucose metabolism parameters were observed from baseline to week 72 with semaglutide 7·2 mg compared with placebo, including HbA_{1c}, fasting plasma glucose, and fasting serum insulin (table 2). Similar improvements were observed in HbA_{1c} (ETD -0·41 percentage points [95% CI -0·45 to -0·37]; p<0·0001), fasting plasma glucose (ETD -0·62 mmol/L [-0·71 to -0·53]; p<0·0001), and fasting serum insulin (ratio to baseline 0·62 [0·56 to 0·68]; p<0·0001) with the trial product estimand (appendix 3 pp 17–19). A change over time graph for HbA_{1c} is shown in figure 3B. When comparing semaglutide 7·2 mg with semaglutide 2·4 mg, significant improvements were only observed in fasting serum insulin (table 2).

During the in-trial period, a higher proportion of participants who had prediabetes at baseline had normoglycaemia at week 72 with semaglutide 7·2 mg

than with semaglutide 2·4 mg and placebo (figure 3). No participants in the semaglutide 7·2 mg group progressed to incident type 2 diabetes, compared with one participant in the semaglutide 2·4 mg group and five participants in the placebo group.

Most participants who were receiving lipid-lowering or antihypertensive medications at baseline had no changes to their medications over the 72 weeks across all groups.

There was no effect of semaglutide 7·2 mg and 2·4 mg on aspartate aminotransferase and alanine aminotransferase concentrations (appendix 3 p 33).

During the in-trial period, the mean change from baseline to week 72 in the number of repetitions in the 30-s sit-to-stand test was 2·73 repetitions with semaglutide 7·2 mg, 2·82 repetitions with semaglutide 2·4 mg, and 2·15 repetitions with placebo (table 2). No significant differences were observed between treatment groups.

Adverse events were reported by 878 (87·5%) of 1004 participants receiving semaglutide 7·2 mg, 169 (84·1%) of 201 receiving semaglutide 2·4 mg, and 156 (77·6%) of 201 receiving placebo (table 3). Most events were mild to moderate in severity and were resolved without permanent discontinuation of treatment. A summary of adverse events by system organ class is described in appendix 3 (pp 20–22).

Serious adverse events were reported by 68 (6·8%) of 1004 participants receiving semaglutide 7·2 mg, 22 (10·9%) of 201 receiving semaglutide 2·4 mg, and 11 (5·5%) of 201 receiving placebo; a list of serious adverse events by system organ class is provided in appendix 3 (p 23). No fatal adverse events were reported in any group.

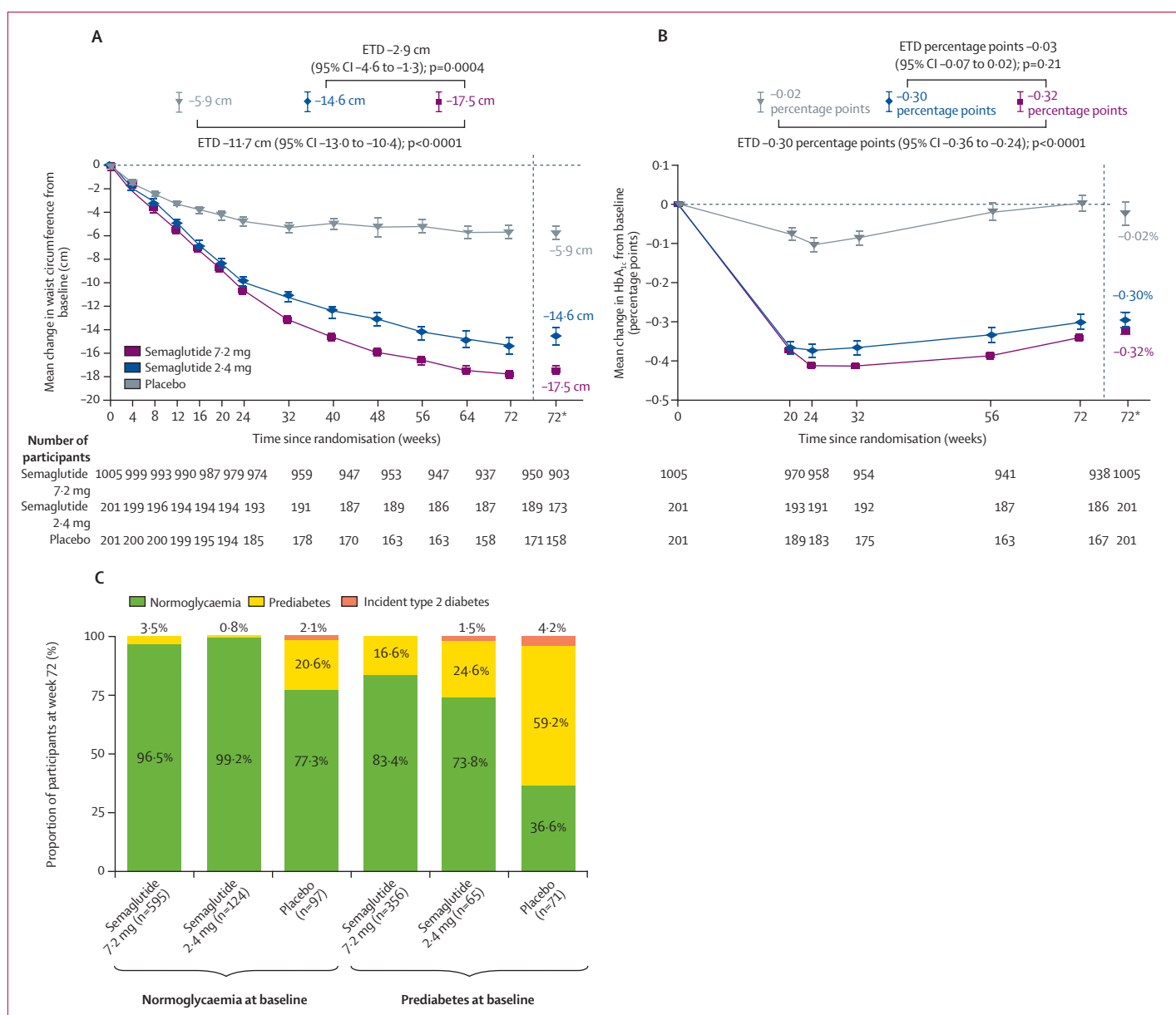


Figure 3: Change in waist circumference and glycaemia from baseline to week 72

Data are from the in-trial observation period and for the full analysis set. Panels A and B show mean change from baseline to week 72 in waist circumference and HbA_{1c}. Panel C shows the shift in glycaemic control for each treatment group (ie, the proportion of participants with normoglycaemia or prediabetes at baseline who had normoglycaemia, prediabetes, or incident type 2 diabetes at week 72). The investigator evaluated glycaemic status periodically during the trial (as detailed in appendix 3 pp 12–13) on the basis of all available relevant information, including medical records, concomitant medication, blood glucose parameters (HbA_{1c} and fasting plasma glucose) and adverse events. The participant's glycaemic status was categorised as normoglycaemia (HbA_{1c} <5.7%), prediabetes (≥5.7% to <6.5%), or type 2 diabetes (HbA_{1c} ≥6.5%) according to the American Diabetes Association's definitions. ETD=estimated treatment difference. *Estimated mean change at week 72.

Gastrointestinal disorders were the most frequently reported adverse events in all treatment groups, occurring in 711 (70.8%) of 1004 participants in the semaglutide 7.2 mg group, 123 (61.2%) of 201 participants in the semaglutide 2.4 mg group, and 86 (42.8%) of 201 participants in the placebo group (table 3). Most events were mild to moderate in severity, of a transitory nature, and occurred mainly during the dose escalation period. Nausea, diarrhoea, vomiting, and constipation

were the most frequently reported gastrointestinal disorders in both semaglutide groups. Time to onset of gastrointestinal disorders and a prevalence plot for these gastrointestinal disorders by severity and treatment group are shown in appendix 3 (pp 34–36). There were more vomiting adverse events in the semaglutide 7.2 mg group than in the semaglutide 2.4 mg and placebo groups, but most of these occurred before increasing the dose to 7.2 mg.

	Semaglutide 7.2 mg (n=1004)			Semaglutide 2.4 mg (n=201)			Placebo (n=201)			Relative risk ratio for semaglutide 7.2 mg vs specified comparator (95% CI)		Absolute risk difference for semaglutide 7.2 mg vs specified comparator (95% CI)	
	n (%)	Events	Rate	n (%)	Event	Rate	n (%)	Events	Rate	vs placebo	vs 2.4 mg	vs placebo	vs 2.4 mg
Adverse events	878 (87.5%)	6363	433.8	169 (84.1%)	1126	386.4	156 (77.6%)	710	266.8	1.13 (1.05 to 1.23)	1.04 (0.98 to 1.12)	0.10 (0.04 to 0.16)	0.03 (-0.02 to 0.09)
Mild	805 (80.2%)	4442	302.9	154 (76.6%)	785	269.4	136 (67.7%)	512	192.4	1.19 (1.08 to 1.32)	1.05 (0.97 to 1.15)	0.13 (0.06 to 0.20)	0.04 (-0.02 to 0.10)
Moderate	506 (50.4%)	1770	120.7	97 (48.3%)	300	102.9	73 (36.3%)	183	68.8	1.39 (1.15 to 1.70)	1.04 (0.90 to 1.23)	0.14 (0.07 to 0.21)	0.02 (-0.05 to 0.10)
Severe	81 (8.1%)	150	10.2	21 (10.4%)	41	14.1	9 (4.5%)	15	5.6	1.80 (0.94 to 3.51)	0.77 (0.50 to 1.22)	0.04 (-0.01 to 0.06)	-0.02 (-0.08 to 0.02)
Serious adverse events	68 (6.8%)	91	6.2	22 (10.9%)	30	10.3	11 (5.5%)	16	6.0	1.24 (0.68 to 2.29)	0.62 (0.40 to 0.98)	0.01 (-0.03 to 0.04)	-0.04 (-0.09 to -0.00)
Adverse events leading to dose reduction	186 (18.5%)	386	26.3	25 (12.4%)	48	16.5	1 (0.5%)	1	0.4	37.24 (6.68 to 211.58)	1.49 (1.02 to 2.21)	0.18 (0.15 to 0.21)	0.06 (0.00 to 0.11)
Adverse events leading to permanent discontinuation	54 (5.4%)	79	5.4	8 (4.0%)	10	3.4	2 (1.0%)	2	0.8	5.41 (1.48 to 20.08)	1.35 (0.67 to 2.77)	0.04 (0.02 to 0.06)	0.01 (-0.03 to 0.04)
Adverse events leading to temporary discontinuation	108 (10.8%)	233	15.9	18 (9.0%)	35	12.0	8 (4.0%)	11	4.1	..	..	..	..
Fatal adverse events*	0	..	..	0	..	..	0	..	..	..	..	..	..
Adverse events of special interest													
Gastrointestinal adverse events (on-treatment)	711 (70.8%)	3059	208.6	123 (61.2%)	479	164.4	86 (42.8%)	218	81.9	1.66 (1.42 to 1.97)	1.16 (1.04 to 1.31)	0.28 (0.21 to 0.35)	0.10 (0.03 to 0.17)
Dysaesthesia (on-treatment)†	230 (22.9%)	340	23.2	12 (6.0%)	17	5.8	1 (0.5%)	1	0.4	46.05 (8.26 to 261.45)	3.84 (2.23 to 6.72)	0.22 (0.19 to 0.25)	0.17 (0.12 to 0.21)
Cardiovascular disorders (on-treatment)	75 (7.5%)	111	7.6	22 (10.9%)	39	13.4	17 (8.5%)	20	7.5	0.88 (0.54 to 1.46)	0.68 (0.44 to 1.07)	-0.01 (-0.06 to 0.03)	-0.04 (-0.09 to 0.01)
Psychiatric disorder adverse events (on-treatment)	69 (6.9%)	91	6.2	16 (8.0%)	20	6.9	15 (7.5%)	18	6.8	0.92 (0.55 to 1.58)	0.86 (0.52 to 1.46)	-0.01 (-0.05 to 0.03)	-0.01 (-0.06 to 0.02)
Neoplasms (in-trial)	56 (5.6%)	67	4.4	11 (5.5%)	16	5.3	10 (5.0%)	14	4.8	1.12 (0.59 to 2.15)	1.02 (0.55 to 1.90)	0.01 (-0.04 to 0.03)	0.00 (-0.04 to 0.03)
Allergic reactions (on-treatment)	49 (4.9%)	56	3.8	12 (6.0%)	12	4.1	7 (3.5%)	7	2.6	1.40 (0.66 to 3.01)	0.82 (0.45 to 1.50)	0.01 (-0.02 to 0.04)	-0.01 (-0.05 to 0.02)
Gallbladder-related disorders (on-treatment)	29 (2.9%)	34	2.3	6 (3.0%)	6	2.1	1 (0.5%)	1	0.4	5.81 (1.02 to 33.66)	0.97 (0.42 to 2.26)	0.02 (0.00 to 0.04)	0.00 (-0.04 to 0.02)
Hepatic disorders (on-treatment)	26 (2.6%)	35	2.4	6 (3.0%)	8	2.7	2 (1.0%)	2	0.8	2.60 (0.70 to 9.88)	0.87 (0.37 to 2.04)	0.02 (-0.01 to 0.03)	0.00 (-0.04 to 0.02)
Injection-site reactions (on-treatment)	25 (2.5%)	66	4.5	2 (1.0%)	2	0.7	3 (1.5%)	8	3.0	1.67 (0.55 to 5.18)	2.50 (0.67 to 9.52)	0.01 (-0.02 to 0.03)	0.02 (-0.01 to 0.03)
Medication error, abuse or misuse, overdose (on-treatment)	25 (2.5%)	30	2.0	8 (4.0%)	10	3.4	7 (3.5%)	7	2.6	0.71 (0.32 to 1.60)	0.63 (0.29 to 1.35)	-0.01 (-0.05 to 0.01)	-0.02 (-0.05 to 0.01)
Malignant neoplasms (in-trial)	10 (1.0%)	11	0.7	2 (1.0%)	2	0.7	2 (1.0%)	2	0.7	1.00 (0.25 to 4.05)	1.00 (0.25 to 4.05)	0.00 (-0.03 to 0.01)	0.00 (-0.03 to 0.01)
Rare events (on-treatment)	10 (1.0%)	11	0.7	0	..	..	1 (0.5%)	1	0.4	2.00 (0.33 to 12.12)	..	0.01 (-0.02 to 0.01)	0.01 (-0.01 to 0.02)
Kidney disorders (on-treatment)	5 (0.5%)	5	0.3	1 (0.5%)	1	0.3	1 (0.5%)	1	0.4	1.00 (0.16 to 6.46)	1.00 (0.16 to 6.46)	0.00 (-0.02 to 0.01)	0.00 (-0.02 to 0.01)
Acute pancreatitis (on-treatment)	3 (0.3%)	3	0.2	0	..	..	0	..	..	..	..	0.00 (-0.02 to 0.01)	0.00 (-0.02 to 0.01)
Hypoglycaemia (on-treatment)	3 (0.3%)	3	0.2	2 (1.0%)	3	1.0	0	..	..	..	0.30 (0.06 to 1.50)	0.00 (-0.02 to 0.01)	-0.01 (-0.03 to 0.00)
Intestinal obstruction (on-treatment)	1 (<0.1%)	1	<0.1	0	..	..	0	..	..	..	..	0.00 (-0.02 to 0.01)	0.00 (-0.02 to 0.01)

Data are from the on-treatment observation period, unless otherwise specified, and for the safety analysis set. Rate refers to events per 100 participant-years. *During the in-trial observation period. †Includes allodynia, burning sensation, dysaesthesia, hyperaesthesia, pain of skin, paraesthesia, sensitive skin, skin burning sensation, skin discomfort, and skin sensitisation.

Table 3: Summary of adverse events

Cardiac disorders were reported by 31 (3.1%) of 1004 participants in the semaglutide 7.2 mg group, eight (4.0%) of 201 in the semaglutide 2.4 mg group, and eight (4.0%) of 201 in the placebo group; hepatobiliary disorders were reported by 37 (3.7%) of 1004, eight (4.0%) of 201, and two (1.0%) of 201, respectively; and acute pancreatitis was reported by three (0.3%) of 1004, none, and none, respectively (appendix 3 pp 20–22).

Dysaesthesia-associated adverse events were reported by 230 (22.9%) of 1004 participants receiving semaglutide 7.2 mg, 12 (6.0%) of 201 participants receiving semaglutide 2.4 mg, and one (0.5%) of 201 participants receiving placebo (table 3; appendix 3 p 24). The most common dysaesthesia-associated adverse events were hyperaesthesia, dysaesthesia, and sensitive skin. Dysaesthesia adverse events were mild for most participants who experienced them (177 in the semaglutide 7.2 mg group, nine in the semaglutide 2.4 mg group, and one in the placebo group), and there were no serious events. Most participants recovered from the event by the end of the trial (197 in the semaglutide 7.2 mg group, ten in the semaglutide 2.4 mg group, and one in the placebo group). Overall, 47 participants in the semaglutide 7.2 mg group, two in the semaglutide 2.4 mg group, and none in the placebo group reduced their dose due to dysaesthesia adverse events. Four participants permanently discontinued treatment due to these adverse events in the semaglutide 7.2 mg group, compared with none in the semaglutide 2.4 mg or placebo groups.

Time to onset and a prevalence plot for dysaesthesia-associated adverse events by severity and treatment group are shown in appendix 3 (p 37).

Permanent discontinuation of the trial product occurred in 54 (5.4%) of 1004 participants receiving semaglutide 7.2 mg, eight (4.0%) of 201 receiving semaglutide 2.4 mg, and two (1.0%) of 201 receiving placebo; 33 (3.3%), four (2.0%), and none were due to gastrointestinal disorders, respectively (appendix 3 p 25).

There was a minimal and comparable change in pulse rate of 1.0 beats per min with semaglutide 7.2 mg, 0.4 beats per min with semaglutide 2.4 mg, and –1.6 beats per min with placebo, over the 72 weeks (appendix 3 p 38). A slight increase in pulse rate was observed in the semaglutide groups around the time of dose escalation, before decreasing.

Discussion

In adults with obesity without type 2 diabetes, an increased dose of once-weekly subcutaneous semaglutide 7.2 mg resulted in significant reductions in bodyweight and an improved likelihood of reaching increasing thresholds of bodyweight reduction, including 25% or greater, compared with placebo. Mean bodyweight reductions of 18.7% with the treatment policy estimand and 20.7% with the trial product estimand were observed with semaglutide 7.2 mg. The tripling in semaglutide dose led to an ETD in bodyweight change of –3.1%

(95% CI –4.7 to –1.6) between semaglutide 7.2 mg and 2.4 mg. Nevertheless, the greatest difference between the doses was seen with higher proportions of participants reaching greater magnitudes of bodyweight loss, including 20% or greater and 25% or greater, with semaglutide 7.2 mg versus 2.4 mg. This suggests that the 7.2 mg dose could provide additional benefits for patients who require a higher magnitude of bodyweight loss, and stratification of patients who could benefit from the higher dose will play a large role in the management of obesity. Additionally, at the end of the trial, 43.2% of participants receiving semaglutide 7.2 mg reached a BMI lower than 30 kg/m². Greater improvements were also seen in cardiovascular risk factors, including waist circumference, blood pressure, lipids, and hsCRP, as well as in glucose metabolism parameters, including HbA_{1c} and fasting plasma glucose with semaglutide 7.2 mg compared with placebo, and modest further improvements were observed with semaglutide 7.2 mg compared with 2.4 mg. The safety and tolerability of semaglutide 7.2 mg were as expected for the GLP-1 receptor agonist class. Gastrointestinal and dysaesthesia adverse events were more common with the 7.2 mg dose than with 2.4 mg and placebo; most were mild to moderate in severity.

The landscape for weight management is continuously changing, and includes monotherapies and a dual-agonist therapy, among others. For people with obesity who are inadequately treated, the greater magnitude of bodyweight change with semaglutide 7.2 mg, compared with lower doses, could help with managing obesity-related complications, which had not been addressed with previous treatments. At the end of treatment in the current study, nearly half of participants in the semaglutide 7.2 mg group had 20% or greater bodyweight reduction and nearly a third had bodyweight reduction of 25% or greater. These levels of bodyweight reduction had positive impacts on the reversion from prediabetes to normoglycaemia, as well as improvements in cardiovascular risk factors. Few participants in all treatment groups were evaluated by the investigators to have progressed to incident type 2 diabetes on the basis of available information (no participants on semaglutide 7.2 mg, one on semaglutide 2.4 mg, and five on placebo). It has previously been shown that bodyweight reductions of 20% or less lead to improvements in obesity-related complications and clinical benefits, including improvements in cardiovascular disease, knee osteoarthritis, obstructive sleep apnoea, and in quality of life, all of which are often a goal of obesity treatment.^{3,4,28–30} Bodyweight reduction had not plateaued at week 72 with semaglutide 7.2 mg, so further reductions might have occurred with a longer trial; long-term extensions to future studies would be advisable to capture the bodyweight loss nadir. Compared with semaglutide 2.4 mg, the rate of bodyweight reduction appeared to increase at a greater extent with the 7.2 mg dose around the time of the final dose escalation from 2.4 mg to

7.2 mg at week 20. Additionally, in a post-hoc analysis, baseline BMI, age, region, and glycaemic status did not appear to affect the magnitude of bodyweight reductions between semaglutide 7.2 mg and 2.4 mg or placebo.

The extent of bodyweight reduction observed with semaglutide 7.2 mg was similar to that observed with the currently approved dual-agonist therapy.¹⁹ From a clinical perspective, the bodyweight reduction observed in the semaglutide 7.2 mg treatment group (18.7%) was relatively similar to that in the SURMOUNT-1 trial, in which participants treated with the highest dose of tirzepatide, 15 mg once weekly, had an observed bodyweight reduction of 21% (both in an assessment of efficacy regardless of treatment discontinuation).¹⁹ In both trials, 91% of participants reached bodyweight reductions of 5% or greater and 31% reached a bodyweight reduction of 25% or greater with semaglutide 7.2 mg while 36% reached a bodyweight reduction of 25% or greater with tirzepatide 15 mg.¹⁹ In the SURMOUNT-5 trial, 32% of participants reached a bodyweight reduction of 25% or greater with tirzepatide 10 mg or 15 mg. However, given salient differences in the populations across these trials (eg, SURMOUNT-1 included participants with BMI ≥ 27 kg/m² and one or more obesity-related complications), such cross-trial comparisons should be interpreted with caution. The increased magnitudes of categorical bodyweight loss seen with these emerging therapies for obesity bring them closer to the magnitudes observed with bariatric surgery,³¹ and could provide an option for pharmacotherapy for those who require substantial bodyweight reduction, without the risks associated with surgery.

The bodyweight-reduction benefits of the increased dose of semaglutide translated to improvements in cardiometabolic risk factors, including blood pressure, lipids, hsCRP, HbA_{1c}, and fasting plasma glucose compared with placebo, and modest further improvements when compared with semaglutide 2.4 mg. Significant improvements, including in waist circumference and lipids (specifically, VLDL cholesterol and triglycerides), were seen with semaglutide 7.2 mg versus 2.4 mg. Notably, by the end of treatment, 83.4% of participants with prediabetes at baseline had reached normoglycaemia with semaglutide 7.2 mg, almost 10% higher than with semaglutide 2.4 mg (73.8%). The goal of chronic weight management is to improve health and improve quality of life and wellbeing.²⁻⁵ As such, increased doses of semaglutide above 2.4 mg might be warranted for those who have not reached their treatment goals with semaglutide 2.4 mg.

Changes in body composition occur with bodyweight loss regardless of the type of intervention, including changes in fat and lean mass.³²⁻³⁴ In the present study, although some loss of lean body volume in the MRI subpopulation was observed, a substantial reduction in total fat volume of 25% was seen with pooled semaglutide,

indicating overall favourable changes in body composition.

Gastrointestinal adverse events were more common with the 7.2 mg versus the 2.4 mg dose (70.8% vs 61.2%), as were dysaesthesia events (eg, tingling, numbness, and burning sensation; 22.9% vs 6.0%). Despite a higher number of these mild-to-moderate adverse events, the percentage of serious adverse events was lower with semaglutide 7.2 mg than with semaglutide 2.4 mg. Notably, tripling the dose of semaglutide from 2.4 mg to 7.2 mg was well tolerated, despite the absence of slow titration, with only relatively small increases in gastrointestinal adverse events compared with semaglutide 2.4 mg, which subsided after titration. The proportion of participants with adverse events leading to permanent discontinuation of the trial product was comparable between the two active groups (5.4% with semaglutide 7.2 mg, 4.0% with semaglutide 2.4 mg, and 1.0% with placebo), supporting a comparable overall benefit–risk ratio. A large proportion of participants were compliant with the highest dose of semaglutide, and the adherence rate remained high in all groups. Adverse events leading to dose reductions were more frequently reported with semaglutide 7.2 mg than with semaglutide 2.4 mg or placebo, driven mainly by gastrointestinal disorders and events related to dysaesthesia.

A higher frequency of dysaesthesia adverse events, which are an emerging side-effect of incretin therapies, was observed with semaglutide 7.2 mg compared with other treatment groups.^{17,18,22,35-38} The mechanism underlying dysaesthesia is unclear; however, given the rapid onset of these events around the time of dose escalation to 7.2 mg, it is not thought to be related to bodyweight loss. Dysaesthesia has been reported with other incretin therapies,^{19,22,37,38} and this adverse effect warrants further investigation. Most events occurred following dose escalation to 7.2 mg, and after escalation the prevalence reduced. Most events were mild in severity and resolved by week 72 without permanent discontinuation. Approximately one in five participants who had dysaesthesia required a dose reduction in the semaglutide 7.2 mg group due to this adverse event. As participants were required to re-consent during the trial following an addition of dysaesthesia adverse events to the investigators brochure and informed consent forms, this could have introduced bias in all treatment groups.

Evaluation of other events in the safety focus areas revealed no unexpected findings and appeared comparable between the three treatment groups. The increased dose of semaglutide did not affect liver enzymes. An increase in pulse rate of around 4–5 beats per min was observed around dose escalation, before stabilising and then decreasing. The increase has been previously documented and is thought to be due to sinoatrial node stimulation and activation of the sympathetic nervous system.³⁹

The STEP UP trial had strengths and limitations. A large number of participants were enrolled in the trial, and a substantial proportion received the highest dose of 7.2 mg. The majority of participants were female, which might have introduced sex-specific biases in the reported data. We acknowledge the limitations in sex and gender reporting, as this trial did not collect data on sex assigned at birth or self-reported gender or sex at baseline. The 5:1:1 randomisation ratio allowed sufficient safety data to be collected for semaglutide 7.2 mg, although a data and safety monitoring board was not required for this trial due to extensive previous research of the molecule; however, monthly laboratory surveillance and safety committee meetings were still held. The trial length was sufficient for investigation; however, a longer follow-up period would have allowed investigation into the bodyweight loss nadir and further insight into the resolution of adverse events with or without discontinuation of the trial product. The coprimary endpoints were based on regulatory requirements to compare semaglutide 7.2 mg with placebo;^{23,24} however, comparisons with semaglutide 2.4 mg were included as confirmatory secondary endpoints. It was also valuable to compare both doses of semaglutide for all endpoints; however, due to the prespecified statistical hierarchy, exploratory post-hoc comparisons were underpowered and therefore not fully representative of the differences between semaglutide 7.2 mg and 2.4 mg.

To conclude, semaglutide 7.2 mg was superior to placebo and semaglutide 2.4 mg in reducing bodyweight, including reaching reductions of 20% or greater and 25% or greater over 72 weeks. A higher proportion of participants with prediabetes reached normoglycaemia with semaglutide 7.2 mg versus semaglutide 2.4 mg and placebo. Metabolic parameters were improved with semaglutide 7.2 mg compared with semaglutide 2.4 mg and placebo. Results from the STEP UP trial therefore support a favourable benefit–risk profile of semaglutide 7.2 mg for weight management in people with obesity, and suggest that a higher dose of semaglutide up to 7.2 mg per week could be used to achieve greater clinical benefits in people not reaching therapeutic goals with once-weekly semaglutide 2.4 mg.

Contributors

SW, PF, JH, IL, JR, and WTG were involved in the conduct of the trial and contributed to data collection. MK, KK, and MQ performed the data analysis. SW, MK, KK, and MQ directly accessed and verified the data. All authors had full access to all the data in the trial and had final responsibility for the decision to submit the manuscript for publication. All authors contributed to data interpretation and writing of the manuscript (assisted by a medical writer paid for by the funder) and vouch for data accuracy and fidelity to the protocol. All authors approved the final version of the manuscript.

Declaration of interests

SW has received payment or honoraria from AstraZeneca, Bausch Health Canada, Boehringer Ingelheim, Eli Lilly, Metsera, Neurogastrix, Novo Nordisk, and Regeneron for academic talks to colleagues; has been provided with medical writing support from AstraZeneca, Boehringer

Ingelheim, Bausch Health Canada, Eli Lilly, and Novo Nordisk; and has an unpaid leadership role in The Obesity Society, and Obesity Canada. PF has received payment or honoraria from Amryt, Bial, Boehringer Ingelheim, FaesFarma, Janssen, Lilly, Merck, Novartis, NovoNordisk, Pronokal, and Sanofi, along with support to attend meetings from Bial, Medinfar, NovoNordisk, and Sanofi. JH has received payment or honoraria from Novo Nordisk, Eli Lilly, VIVUS, and AstraZeneca; and has received fees to participate on a data safety monitoring board or advisory board for Novo Nordisk, Eli Lilly, and VIVUS. MK is a full-time employee of, and holds shares in, Novo Nordisk; and is named on a US patent for semaglutide (“Semaglutide in Medical Therapy”, Patent Number: US 12,029,779 B2). KK is a full-time employee of, and holds shares in, Novo Nordisk. IL has received grants from Boehringer Ingelheim, Dexcom, Eli Lilly, Novo Nordisk, Roche, and Pfizer, with all payments made to her institution; has received payment or honoraria from AbbVie, Altimimmune, Alveus Therapeutics, Amgen, Antag Therapeutics, AstraZeneca, Bayer, Betagenon, Bioio, Biomea, Boehringer Ingelheim, Carmot, Cytokine Pharma, Eli Lilly, Intercept, Janssen/J&J, Juvena, Keros Therapeutic, Mediflix, Merck, Metsera, Neurocrine, Novo Nordisk, Pharmaventures, Pfizer, Regeneron, Roche, Sanofi, Shionogi, Source Bio, Structure Therapeutics, TARGET RWE, Terns Pharma, The Comm Group, Verdiva Bio, WebMD, and Zealand Pharma; and has received support to attend meetings from AstraZeneca, Boehringer Ingelheim, Novo Nordisk, and Eli Lilly. MDLAQ-P is a full-time employee of Novo Nordisk and holds shares in Novo Nordisk, Eli Lilly, and Genmab. JR has received research grants from Amgen, Applied Therapeutics, AstraZeneca, Biomea Fusion, Boehringer Ingelheim, Carmot, Corcept, Eli Lilly, Hanmi, Merck, Novartis, Novo Nordisk, Oramed, Regeneron, Pfizer, Sanofi, Structure Therapeutics and Terns; and has served on scientific advisory boards and received honorarium or consulting fees from Amgen, Applied Therapeutics, Biomea Fusion, Eli Lilly, Hanmi, Novo Nordisk, Oramed, Regeneron, Regor, Roche, Sanofi, Structure Therapeutics, and Terns; and received honoraria for lectures from Eli Lilly, Novo Nordisk, and Sanofi. WTG has received grants as a principal investigator from Novo Nordisk, Eli Lilly & Co, Pfizer, Epitomee, Neurovalens, Zealand, Carmot/Roche, Terns Pharmaceuticals, Viking Therapeutics, and Pfizer, and for research from the UAB Diabetes Research Center (DK079626) funded by the US National Institutes of Health and National Institute of Diabetes and Digestive and Kidney Diseases, with all payments made to his institution; has received consulting fees as an advisory board member from Boehringer Ingelheim, Eli Lilly, Novo Nordisk, Pfizer, Fractyl Health, Alnylam Pharmaceuticals, Inogen, Zealand, Allurion, Carmot/Roche, Terns Pharmaceuticals, Neurocrine, Keros Therapeutics, Gan & Lee, NodThera, and Regeneron; has received fees to participate on a data safety monitoring board or advisory board for Boehringer Ingelheim and Eli Lilly; and is an unpaid member of the *The Lancet Diabetes & Endocrinology* Commission on definition and diagnostic criteria of clinical obesity, American Association of Clinical Endocrinology (AACE) update of obesity treatment algorithm, American Diabetes Association (ADA) nutrition recommendations update, and ADA Standards of Care for Obesity.

Data sharing

Data will be shared with bona fide researchers who submit a research proposal approved by the independent review board. Individual participant data will be shared in datasets in a de-identified and anonymised format. Information about data access request proposals can be found at <https://www.novonordisk-trials.com>.

Acknowledgments

This trial was sponsored by Novo Nordisk and is registered with ClinicalTrials.gov (NCT05646706). We thank the trial participants, investigators, and trial site staff who conducted the trial. We also thank Kristoffer Jensen Kolnes for his input with reviewing the manuscript. Medical writing support was provided by Abbie Richold, of Apollo, OPEN Health Communications, and funded by Novo Nordisk, in accordance with Good Publication Practice guidelines. During the preparation of this work the authors used LÉON to produce German and Greek translations of the Abstract. After using this tool, the contents were checked for accuracy by native speakers and the authors take full responsibility for the final version.

References

- Phelps NH, Singleton RK, Zhou B, et al, and the NCD Risk Factor Collaboration (NCD-RisC). Worldwide trends in underweight and obesity from 1990 to 2022: a pooled analysis of 3663 population-representative studies with 222 million children, adolescents, and adults. *Lancet* 2024; **403**: 1027–50.
- Kim KK, Haam JH, Kim BT, et al, and the Committee of Clinical Practice Guidelines, Korean Society for the Study of Obesity (KSSO). Evaluation and treatment of obesity and its comorbidities: 2022 update of clinical practice guidelines for obesity by the Korean Society for the Study of Obesity. *J Obes Metab Syndr* 2023; **32**: 1–24.
- Yumuk V, Tsigos C, Fried M, et al, and the Obesity Management Task Force of the European Association for the Study of Obesity. European guidelines for obesity management in adults. *Obes Facts* 2015; **8**: 402–24.
- Garvey WT, Mechanick JL, Brett EM, et al, and the Reviewers of the AACE/ACE Obesity Clinical Practice Guidelines. American Association of Clinical Endocrinologists and American College of Endocrinology Comprehensive Clinical Practice Guidelines for medical care of patients with obesity. *Endocr Pract* 2016; **22** (suppl 3): 1–203.
- Markovic TP, Proietto J, Dixon JB, et al. The Australian obesity management algorithm: a simple tool to guide the management of obesity in primary care. *Obes Res Clin Pract* 2022; **16**: 353–63.
- Wilding JPH, Batterham RL, Calanna S, et al, and the STEP 1 Study Group. Once-weekly semaglutide in adults with overweight or obesity. *N Engl J Med* 2021; **384**: 989–1002.
- Davies M, Færch L, Jeppesen OK, et al, and the STEP 2 Study Group. Semaglutide 2.4 mg once a week in adults with overweight or obesity, and type 2 diabetes (STEP 2): a randomised, double-blind, double-dummy, placebo-controlled, phase 3 trial. *Lancet* 2021; **397**: 971–84.
- Wadden TA, Bailey TS, Billings LK, et al, and the STEP 3 Investigators. Effect of subcutaneous semaglutide vs placebo as an adjunct to intensive behavioral therapy on body weight in adults with overweight or obesity: the STEP 3 randomized clinical trial. *JAMA* 2021; **325**: 1403–13.
- Rubino D, Abrahamsson N, Davies M, et al, and the STEP 4 Investigators. Effect of continued weekly subcutaneous semaglutide vs placebo on weight loss maintenance in adults with overweight or obesity: the STEP 4 randomized clinical trial. *JAMA* 2021; **325**: 1414–25.
- Garvey WT, Batterham RL, Bhatta M, et al, and the STEP 5 Study Group. Two-year effects of semaglutide in adults with overweight or obesity: the STEP 5 trial. *Nat Med* 2022; **28**: 2083–91.
- Kadowaki T, Isendahl J, Khalid U, et al, and the STEP 6 Investigators. Semaglutide once a week in adults with overweight or obesity, with or without type 2 diabetes in an east Asian population (STEP 6): a randomised, double-blind, double-dummy, placebo-controlled, phase 3a trial. *Lancet Diabetes Endocrinol* 2022; **10**: 193–206.
- Mu Y, Bao X, Eliaschewitz FG, et al, and the STEP 7 Study Group. Efficacy and safety of once weekly semaglutide 2.4 mg for weight management in a predominantly east Asian population with overweight or obesity (STEP 7): a double-blind, multicentre, randomised controlled trial. *Lancet Diabetes Endocrinol* 2024; **12**: 184–95.
- Rubino DM, Greenway FL, Khalid U, et al, and the STEP 8 Investigators. Effect of weekly subcutaneous semaglutide vs daily liraglutide on body weight in adults with overweight or obesity without diabetes: the STEP 8 randomized clinical trial. *JAMA* 2022; **327**: 138–50.
- Bliddal H, Bays H, Czernichow S, et al, and the STEP 9 Study Group. Once-weekly semaglutide in persons with obesity and knee osteoarthritis. *N Engl J Med* 2024; **391**: 1573–83.
- McGowan BM, Bruun JM, Capehorn M, et al, and the STEP 10 Study Group. Efficacy and safety of once-weekly semaglutide 2.4 mg versus placebo in people with obesity and prediabetes (STEP 10): a randomised, double-blind, placebo-controlled, multicentre phase 3 trial. *Lancet Diabetes Endocrinol* 2024; **12**: 631–42.
- Kosiborod MN, Abildstrøm SZ, Borlaug BA, et al, and the STEP-HFpEF Trial Committees and Investigators. Semaglutide in patients with heart failure with preserved ejection fraction and obesity. *N Engl J Med* 2023; **389**: 1069–84.
- FDA. Wegovy prescribing information. March 2024. https://www.accessdata.fda.gov/drugsatfda_docs/label/2024/215256s011lbl.pdf (accessed Aug 12, 2025).
- EMA. Wegovy summary of product characteristics. Jan 8, 2025. https://www.ema.europa.eu/en/documents/product-information/wegovy-epar-product-information_en.pdf (accessed Aug 12, 2025).
- Jastreboff AM, Aronne LJ, Ahmad NN, et al, and the SURMOUNT-1 Investigators. Tirzepatide once weekly for the treatment of obesity. *N Engl J Med* 2022; **387**: 205–16.
- Aronne LJ, Horn DB, le Roux CW, et al, and the SURMOUNT-5 Trial Investigators. Tirzepatide as compared with semaglutide for the treatment of Obesity. *N Engl J Med* 2025; **393**: 26–36.
- Strathe A, Horn DB, Larsen MS, et al. A model-based approach to predict individual weight loss with semaglutide in people with overweight or obesity. *Diabetes Obes Metab* 2023; **25**: 3171–80.
- Knop FK, Aroda VR, do Vale RD, et al, and the OASIS 1 Investigators. Oral semaglutide 50 mg taken once per day in adults with overweight or obesity (OASIS 1): a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet* 2023; **402**: 705–19.
- FDA. Guidance for industry: developing products for weight management. Feb 15, 2007. <https://www.regulations.gov/document/FDA-2007-D-0435-0002> (accessed June 17, 2025).
- EMA. Clinical evaluation of medicinal products used in weight control—Scientific guideline. Jan 1, 2017. <https://www.ema.europa.eu/en/clinical-evaluation-medicinal-products-used-weight-control-scientific-guideline> (accessed June 17, 2025).
- McEvoy BW. Missing data in clinical trials for weight management. *J Biopharm Stat* 2016; **26**: 30–36.
- Kushner RF, Calanna S, Davies M, et al. Semaglutide 2.4 mg for the treatment of obesity: key elements of the STEP trials 1 to 5. *Obesity* 2020; **28**: 1050–61.
- Wharton S, Freitas P, Hjelmæsaeth J, et al. Efficacy and safety of semaglutide 7.2 mg in obesity: the STEP UP trial. American Diabetes Association 85th Scientific Sessions; June 20–23, 2025. (abstr #2025-LB-6521).
- Wharton S, Lau DCW, Vallis M, et al. Obesity in adults: a clinical practice guideline. *CMAJ* 2020; **192**: E875–91.
- Horn DB, Almandoz JP, Look M. What is clinically relevant weight loss for your patients and how can it be achieved? A narrative review. *Postgrad Med* 2022; **134**: 359–75.
- Garvey WT. New horizons. A new paradigm for treating to target with second-generation obesity medications. *J Clin Endocrinol Metab* 2022; **107**: e1339–47.
- O'Brien PE, Hindle A, Brennan L, et al. Long-term outcomes after bariatric surgery: a systematic review and meta-analysis of weight loss at 10 or more years for all bariatric procedures and a single-centre review of 20-year outcomes after adjustable gastric banding. *Obes Surg* 2019; **29**: 3–14.
- Karakasis P, Patoulis D, Fragakakis N, Mantzoros CS. Effect of glucagon-like peptide-1 receptor agonists and co-agonists on body composition: systematic review and network meta-analysis. *Metabolism* 2025; **164**: 156113.
- Keenan S, Cooke MB, Belski R. The effects of intermittent fasting combined with resistance training on lean body mass: a systematic review of human studies. *Nutrients* 2020; **12**: 2349.
- Haghighat N, Ashtary-Larky D, Bagheri R, et al. Preservation of fat-free mass in the first year after bariatric surgery: a systematic review and meta-analysis of 122 studies and 10,758 participants. *Surg Obes Relat Dis* 2022; **18**: 964–82.
- Burke OM, Sa B, Cespedes DA, Tosti A. Dermatologic implications of glucagon-like peptide-1 receptor agonist medications. *Skin Appendage Disord* 2025; published online Feb 14. <https://doi.org/10.1159/000544023>.
- Stokes ML, Linnebur SA, Tietz S, Fixen DR. A case of semaglutide-induced skin burning. *Ann Intern Med Clin Cases* 2025; **4**: e240568.
- Jastreboff AM, Kaplan LM, Frias JP, et al, and the Retatrutide Phase 2 Obesity Trial Investigators. Triple-hormone-receptor agonist retatrutide for obesity — a phase 2 trial. *N Engl J Med* 2023; **389**: 514–26.
- FDA. Zepbound prescribing information. December 2024. https://www.accessdata.fda.gov/drugsatfda_docs/label/2024/217806s013lbl.pdf (accessed July 3, 2025).
- Heuvelman VD, Van Raalte DH, Smits MM. Cardiovascular effects of glucagon-like peptide 1 receptor agonists: from mechanistic studies in humans to clinical outcomes. *Cardiovasc Res* 2020; **116**: 916–30.